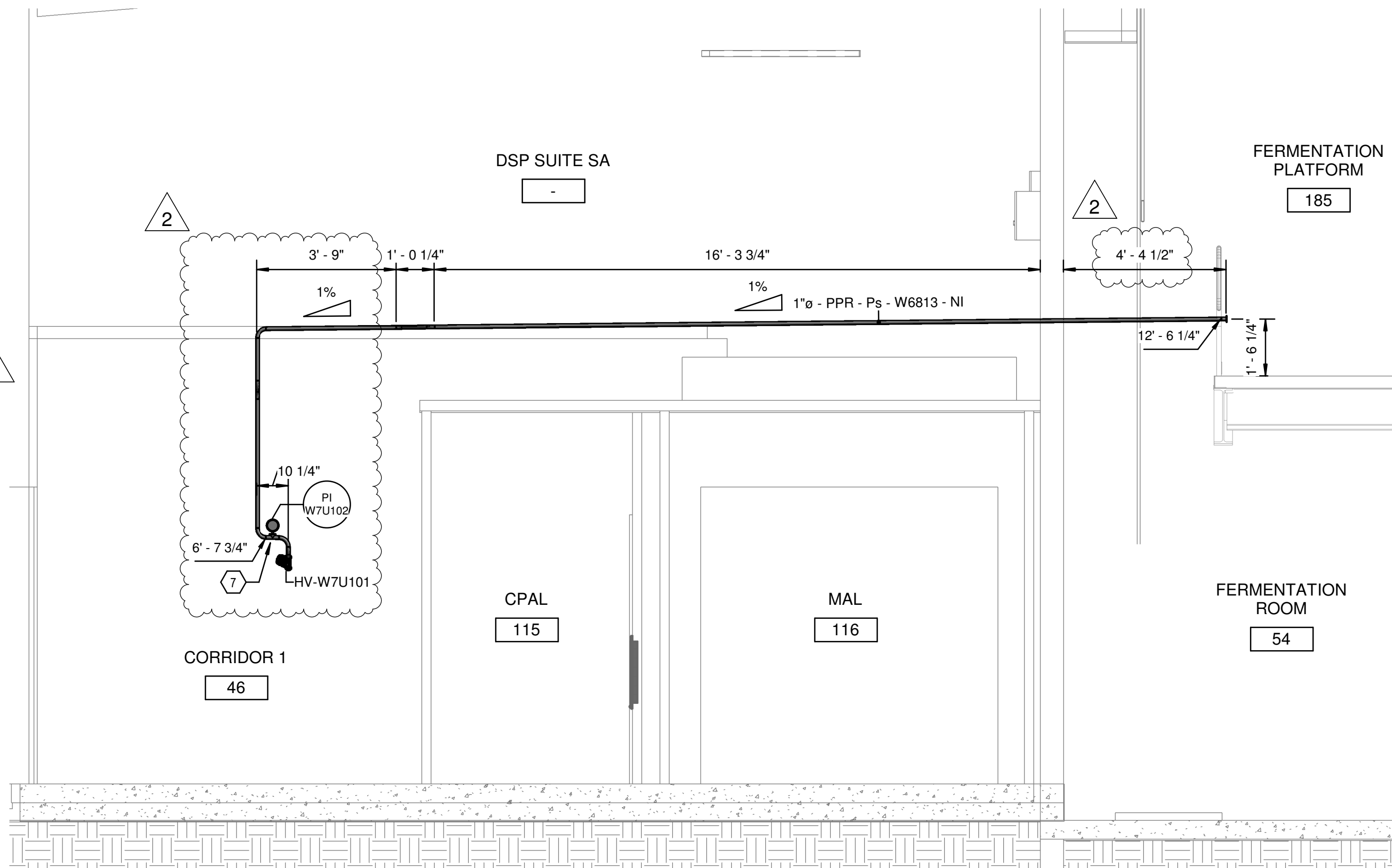
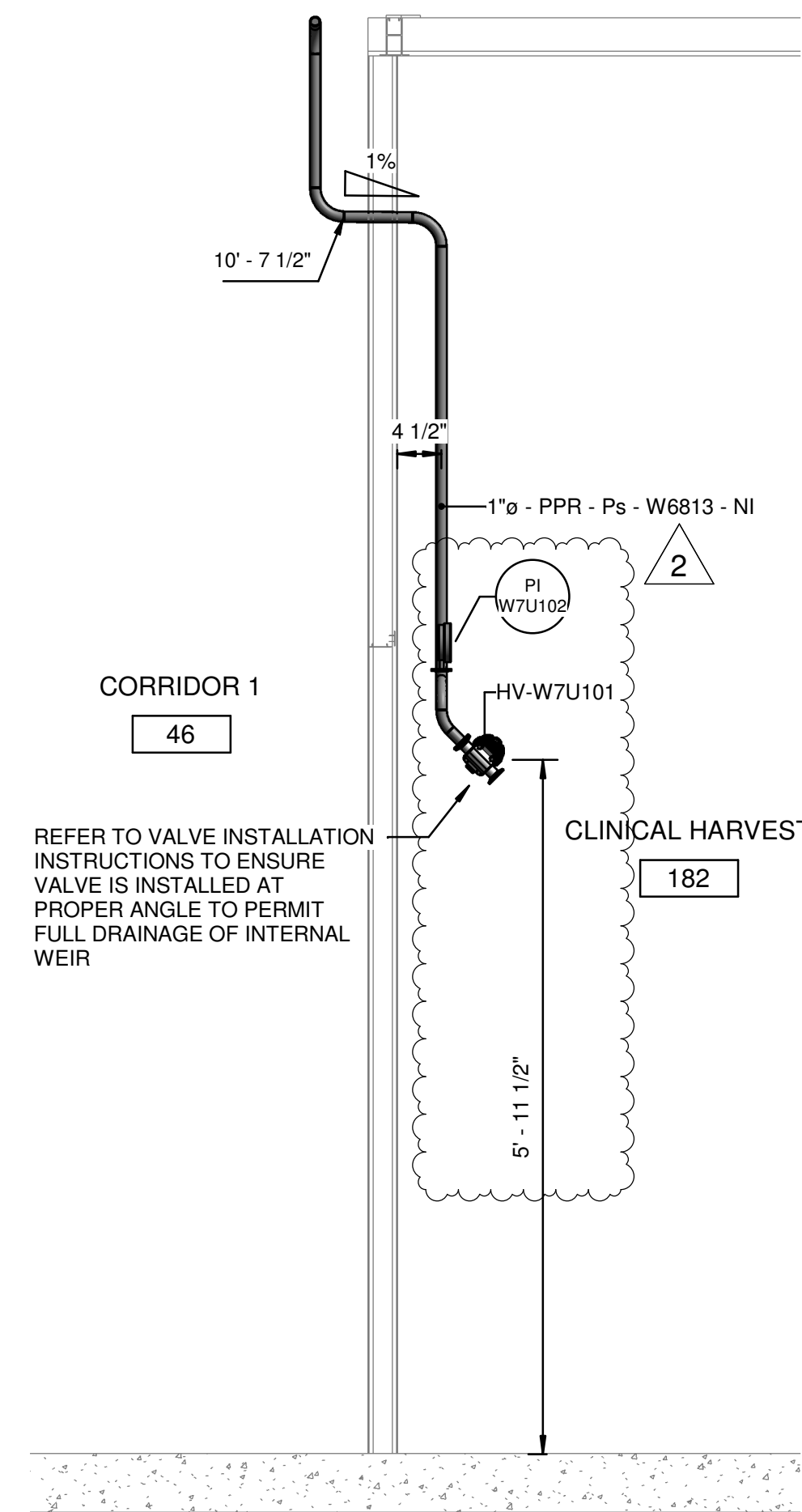


DRAWING NOTES:

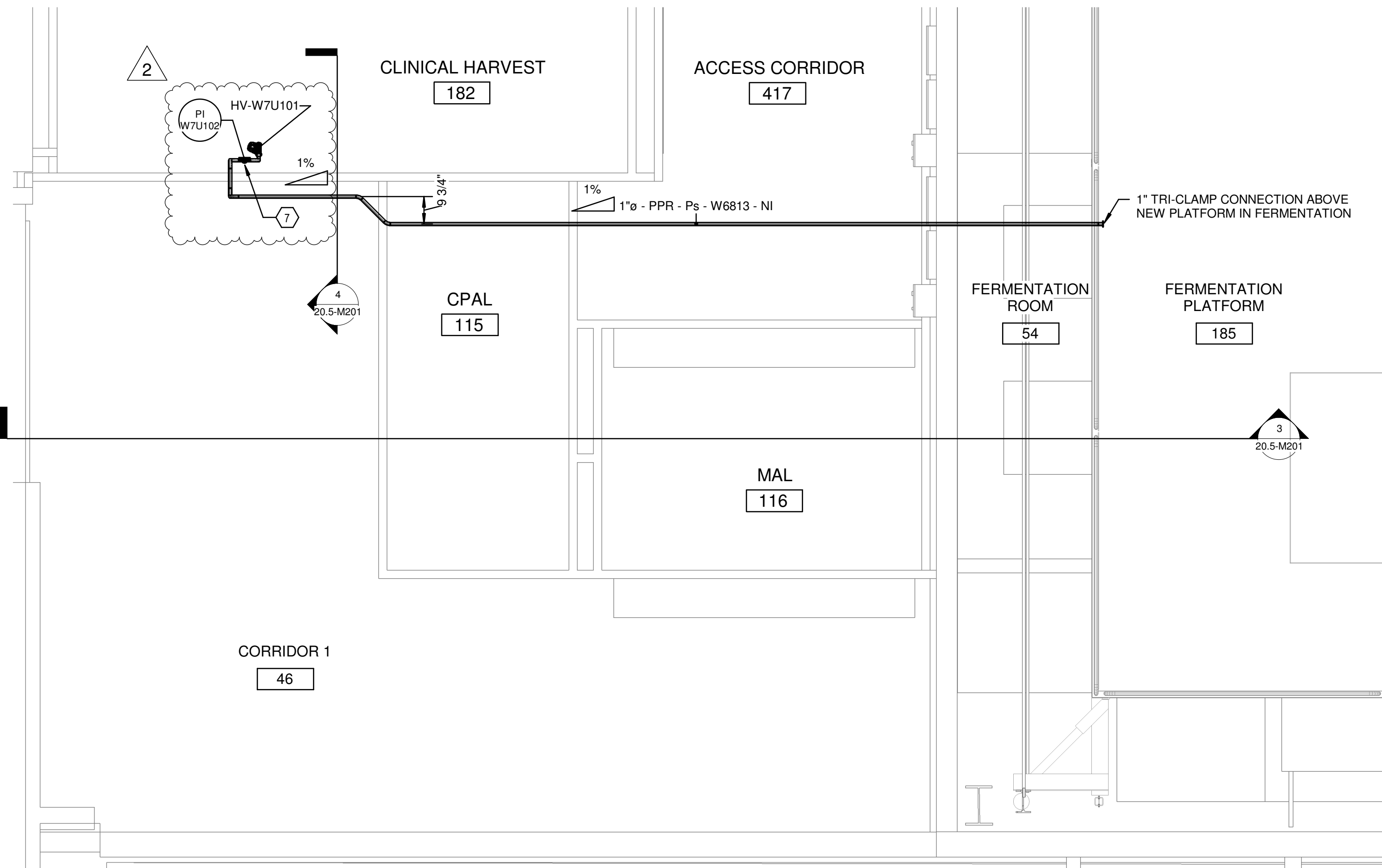
- 1) THIS DRAWING IS INTENDED FOR GENERAL ROUTE/LOCATIONS ONLY. COORDINATE EXACT ROUTE/LOCATIONS ON SITE.
- 2) TRANSLUCENT PIPING/EQUIPMENT REPRESENTS EXISTING PIPING THAT IS TO REMAIN.
- 3) SOLID PIPING/EQUIPMENT WITH BOLD LINES REPRESENTS NEW ELEMENTS.
- 4) COORDINATE WORK ON LINES WITH BIOVECTRA TO ENSURE LINES ARE DE-ENERGIZED WITH APPROPRIATE LOCK-OUT/TAG-OUT ENGAGED.
- 5) UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
- 6) REFER TO DRAWING 20.5-M102 (PID-W7U) FOR MORE DETAILS. 20.5-M102 SUPERCEDES 20.5-M101 PREVIOUSLY ISSUED WITH RFP DOCUMENTS.
- 7) INSTALL SHORT OUTLET TRI-CLAMP CONNECTION TEE IN NOMINALLY HORIZONTAL ORIENTATION FOR PRESSURE GAUGE. PI-W7U102 PROVIDED BY OTHERS.



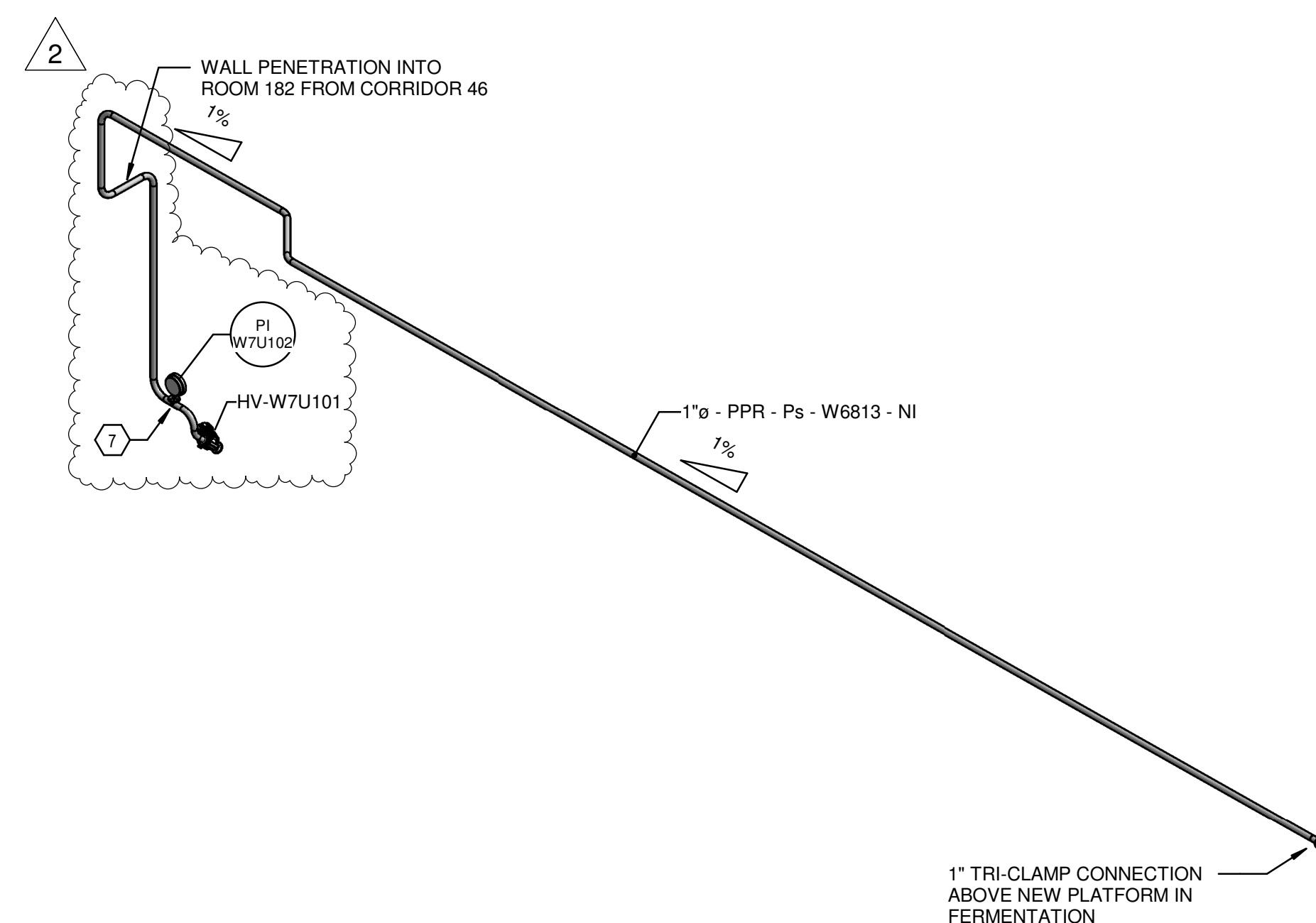
3 FERMENTATION - PROCESS PRODUCT SECTION 1  
3/8" = 1'-0"



4 FERMENTATION - PROCESS PRODUCT SECTION 2  
3/4" = 1'-0"



1 FERMENTATION - PROCESS PRODUCT LAYOUT  
3/8" = 1'-0"



2 FERMENTATION - PROCESS PRODUCT ISOMETRIC

LEGEND:

- DEMOLITION PIPING
- EXISTING PIPING
- XX" - XX - XX  
↑  
↑  
↑  
PIPE TYPE (SEE SPECIFICATION)  
SERVICE (SEE SPECIFICATION)  
PIPE SIZE
- C.O.P. = XX' - XX" CENTER OF PIPE ELEVATION
- XX' - XX" POINT ELEVATION
- XX' - XX' PIPE TYPE CHANGE (SEE SPECIFICATION)
- X% SLOPE INDICATOR
- HP / LP HIGH POINT / LOW POINT

NOTES:

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2. DIMENSIONS NOTED ON LAYOUT ARE TO CENTER OF PIPE.
3. REFER TO PID LEGEND SHEET FOR LIST OF FLUID SERVICE DESIGNATIONS.
4. ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).

|     |                         |          |
|-----|-------------------------|----------|
| 2   | ISSUED FOR CONSTRUCTION | 31-01-22 |
| 1   | ISSUED FOR PRICING      | 29-10-21 |
| NO. | Revision                | DD-MM-YY |

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3. CONTRACTOR TO COORDINATE ALL ROUTING ON SITE PRIOR TO INSTALLATION. ANY ADJUSTMENTS REQUIRED TO ACCOMMODATE EXISTING CONDITIONS SHALL BE INCLUDED IN TENDER.
4. FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

Drawing Title:

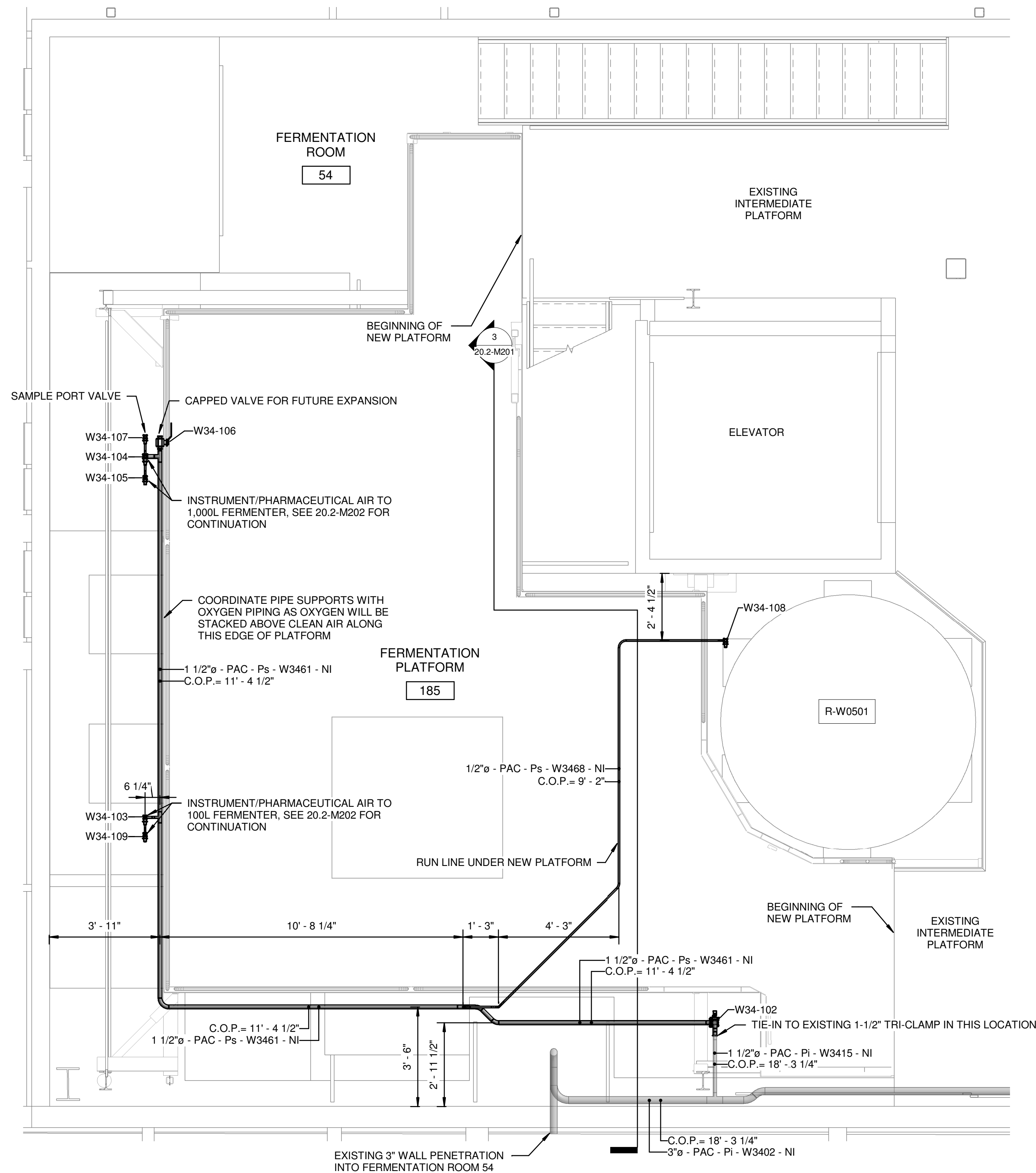
FERMENTATION - PROCESS PRODUCT PLANS

|          |          |                |           |
|----------|----------|----------------|-----------|
| Date:    | 26/10/21 | Project No.:   | 21-515    |
| Scale:   | AS NOTED | DWG No.:       | 20.5-M201 |
| Drawn:   | KJG      | Origin Dwg:    |           |
| Checked: | SKR      | File No.:      | 21-515    |
|          |          | BioVectra Dwg: |           |

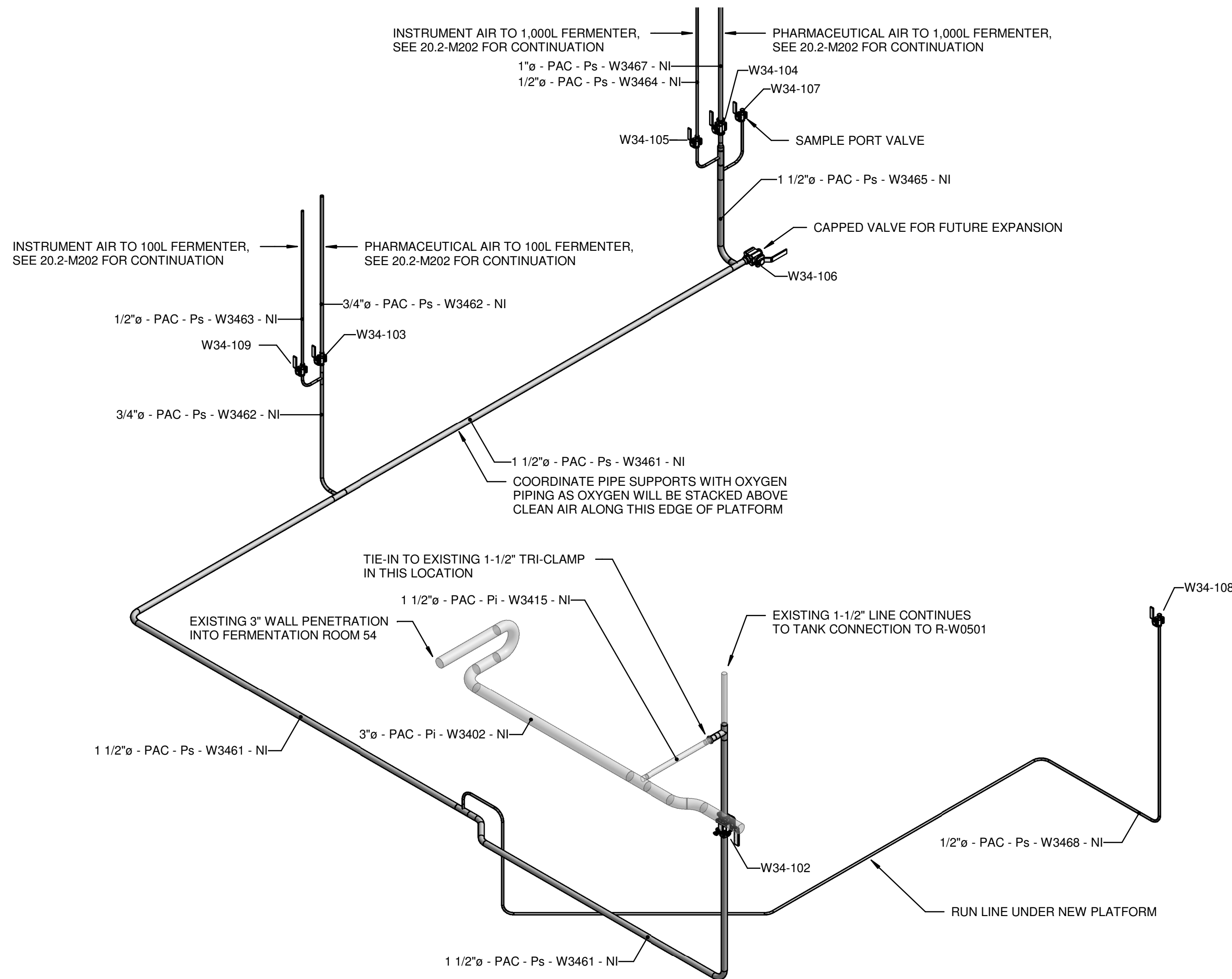


DRAWING NOTES:

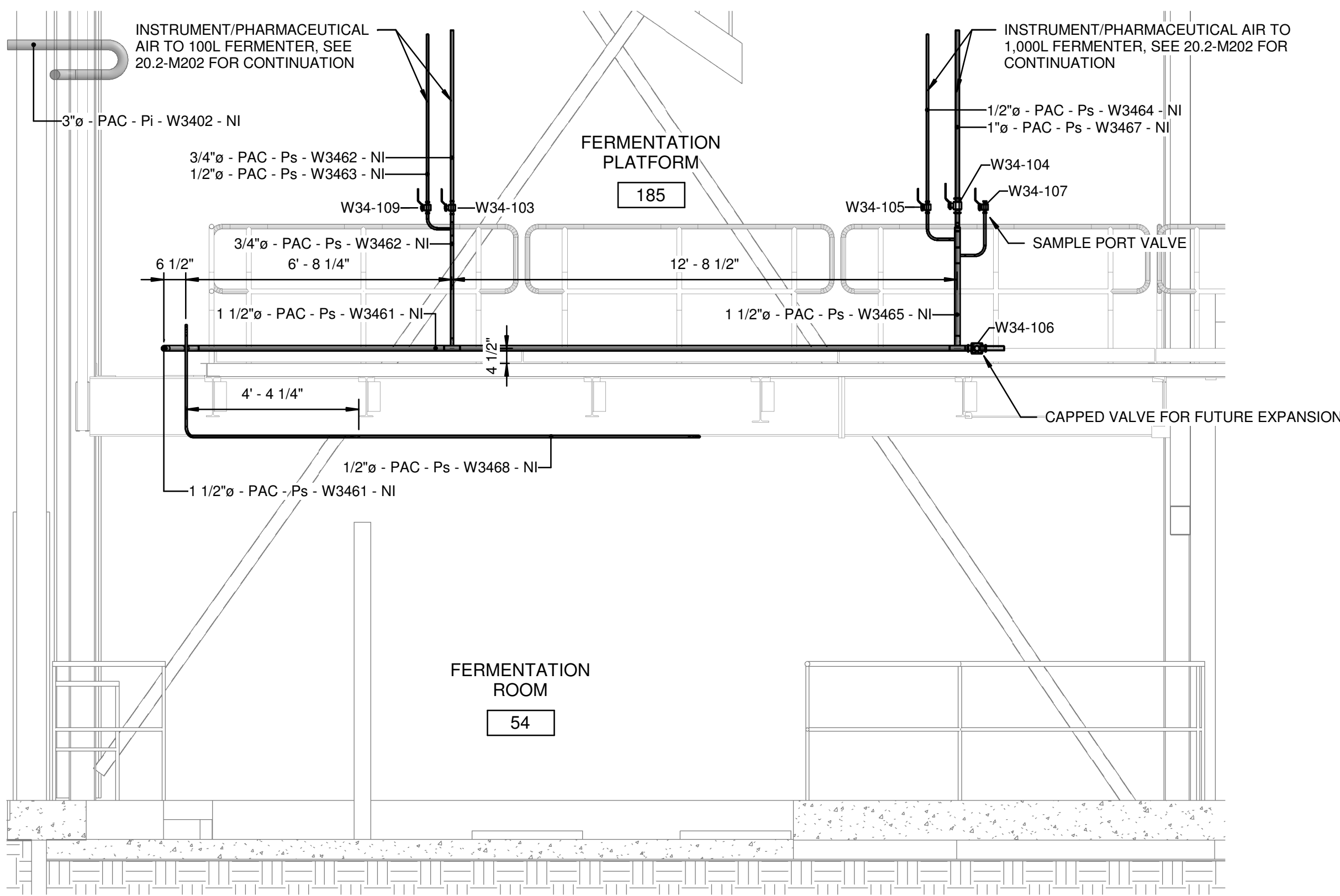
- 1) THIS DRAWING IS INTENDED FOR GENERAL ROUTE/LOCATIONS ONLY, COORDINATE EXACT ROUTE/LOCATIONS ON SITE.
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- 4) COORDINATE WORK ON LINES WITH BIOVECTRA TO ENSURE LINES ARE DE-ENERGIZED WITH APPROPRIATE LOCK-OUT/TAG-OUT ENGAGED.
- 5) ALL SUPPORTS AND ASSOCIATED SUPPORT STRUCTURES ARE BY THIS CONTRACTOR. WHERE REQUIRED TO CREATE SUPPORT STRUCTURES FROM THE FERMENTATION PLATFORM, OR OTHERWISE, STAINLESS HSS SHALL BE USED. LOCATIONS OF SUPPORT STRUCTURE MUST BE COORDINATED WITH THE ENGINEER (MCA).
- 6) UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
- 7) REFER TO DRAWING 20.2-M101 (PID-W34) FOR MORE DETAILS.



1 FERMENTATION - CLEAN AIR LAYOUT  
3/8" = 1'-0"



2 FERMENTATION - CLEAN AIR ISOMETRIC



3 FERMENTATION - CLEAN AIR SECTION  
3/8" = 1'-0"

LEGEND:

- DEMOLITION PIPING
- ===== EXISTING PIPING
- XX" - XX" - XX NEW PIPING
- PIPE TYPE (SEE SPECIFICATION)
- SERVICE (SEE SPECIFICATION)
- PIPE SIZE
- C.O.P. = XX" - XX" CENTER OF PIPE ELEVATION
- XX" - XX" POINT ELEVATION
- XX ← XX PIPE TYPE CHANGE (SEE SPECIFICATION)
- X% SLOPE INDICATOR
- HP / LP HIGH POINT / LOW POINT

NOTES:

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  2. DIMENSIONS NOTED ON LAYOUT ARE TO CENTER OF PIPE.
  3. REFER TO PID LEGEND SHEET FOR LIST OF FLUID SERVICE DESIGNATIONS.
  4. ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).
- | NO. | Revision                 | DD-MM-YY |
|-----|--------------------------|----------|
| 4   | ISSUED FOR CONSTRUCTION  | 31-01-22 |
| 3   | ISSUED FOR 20-CCO-01     | 22-11-21 |
| 2   | ISSUED FOR 20-ADDENDUM-1 | 09-11-21 |
| 1   | ISSUED FOR PRICING       | 29-10-21 |

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4. FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

Drawing Title:

FERMENTATION - CLEAN AIR PLANS

|          |          |                |           |
|----------|----------|----------------|-----------|
| Date:    | 18/10/21 | Project No.:   | 21-515    |
| Scale:   | AS NOTED | DWG No.:       | 20.2-M201 |
| Drawn:   | KJG      | Origin Dwg:    |           |
| Checked: | SKR      | File No.:      | 21-515    |
|          |          | BioVectra Dwg: |           |



DRAWING NOTES:

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- COORDINATE WORK ON LINES WITH BIOVECTRA TO ENSURE LINES ARE DE-ENERGIZED WITH APPROPRIATE LOCK-OUT/TAG-OUT ENGAGED.
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- UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
- REFER TO DRAWING 20.2-M101 (PID-W34) FOR MORE DETAILS.

LEGEND:

|                   |                                         |
|-------------------|-----------------------------------------|
| -----             | DEMOLITION PIPING                       |
| =====             | EXISTING PIPING                         |
| XX" - XX - XX     | NEW PIPING                              |
| ---               | PIPE TYPE (SEE SPECIFICATION)           |
| ---               | SERVICE (SEE SPECIFICATION)             |
| ---               | PIPE SIZE                               |
| C.O.P.= XX' - XX" | CENTER OF PIPE ELEVATION                |
| XX' - XX"         | POINT ELEVATION                         |
| XX' - XX"         | PIPE TYPE CHANGE<br>(SEE SPECIFICATION) |
| X%                | SLOPE INDICATOR                         |
| HP / LP           | HIGH POINT / LOW POINT                  |

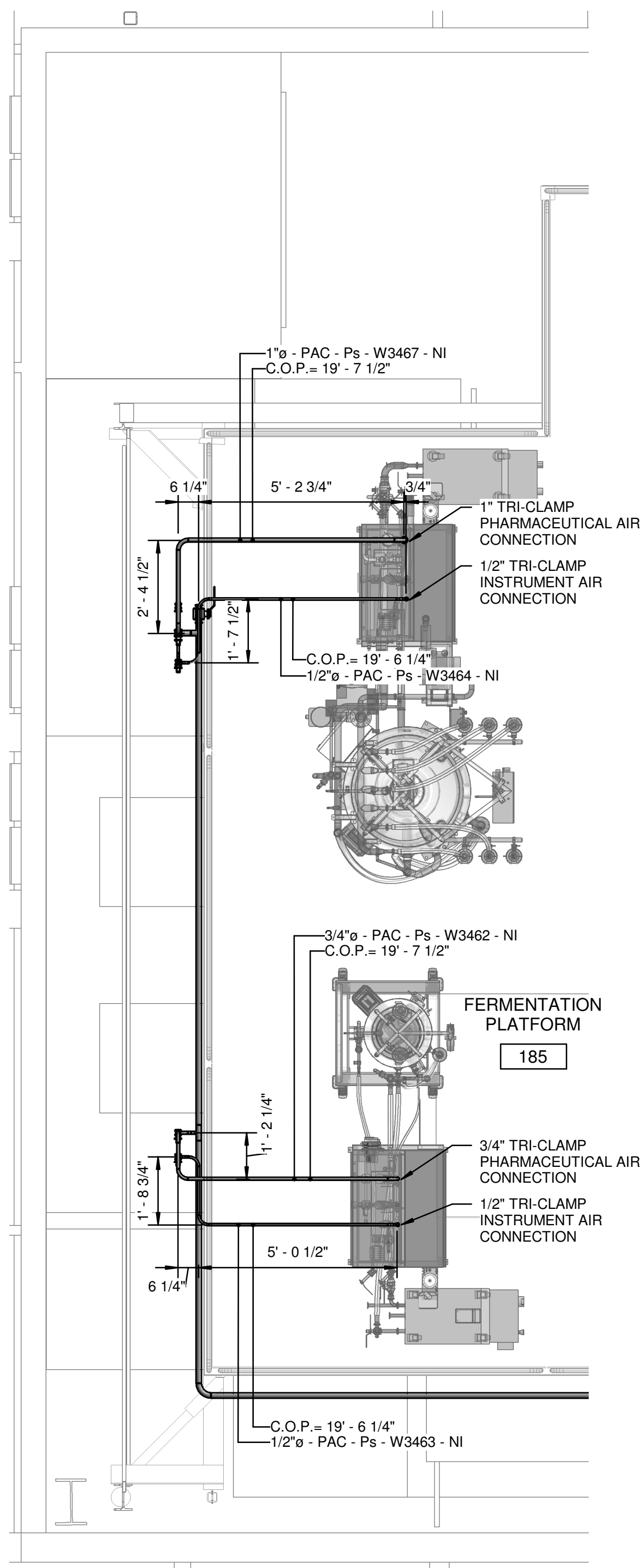
NOTES:

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- ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).

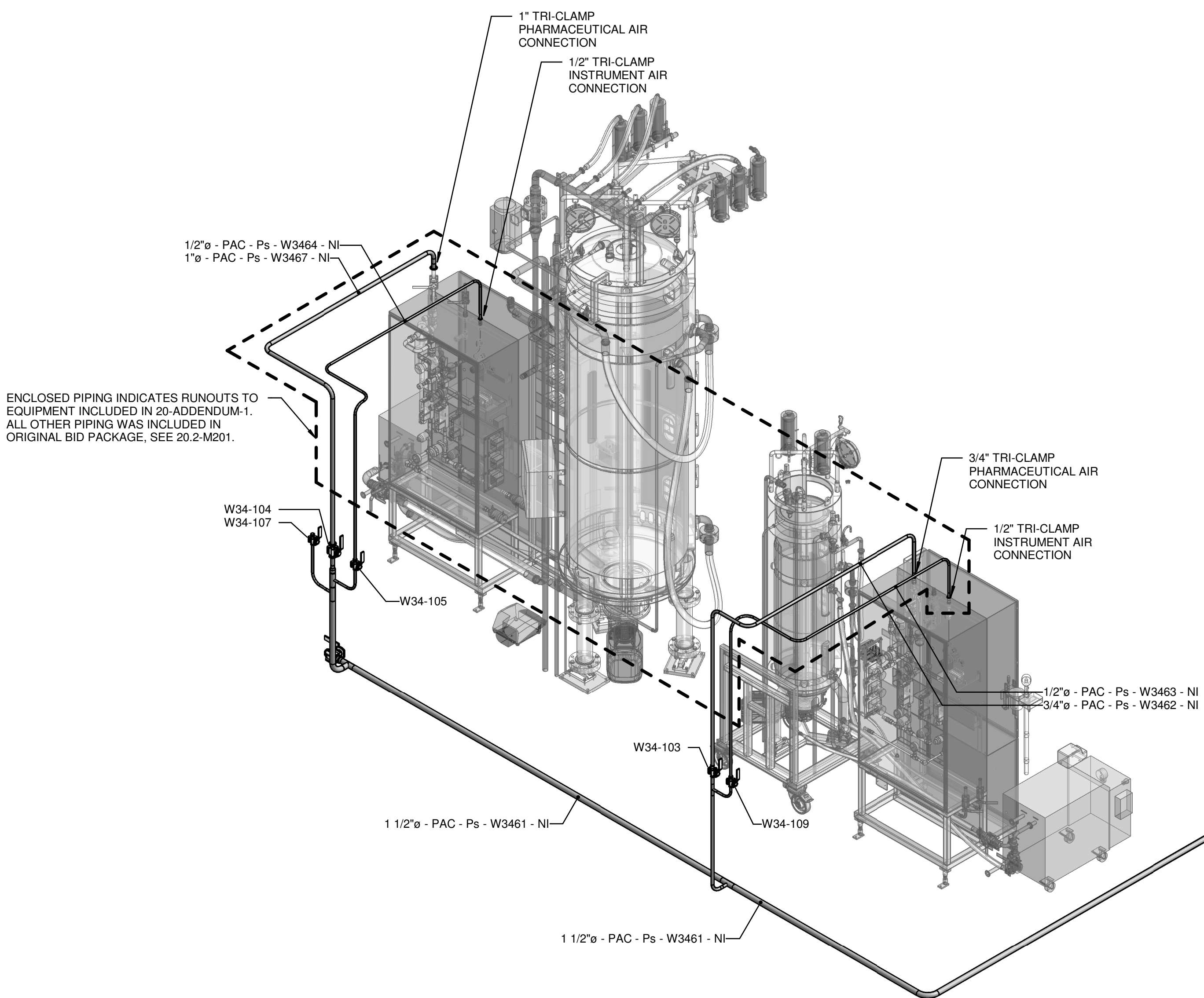
|     |                          |          |
|-----|--------------------------|----------|
| 3   | ISSUED FOR CONSTRUCTION  | 31-01-22 |
| 2   | ISSUED FOR 20-CCO-01     | 22-11-21 |
| 1   | ISSUED FOR 20-ADDENDUM-1 | 09-11-21 |
| NO. | Revision                 | DD-MM-YY |

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1 FERMENTATION - CLEAN AIR RUNOUTS LAYOUT  
3/8" = 1'-0"



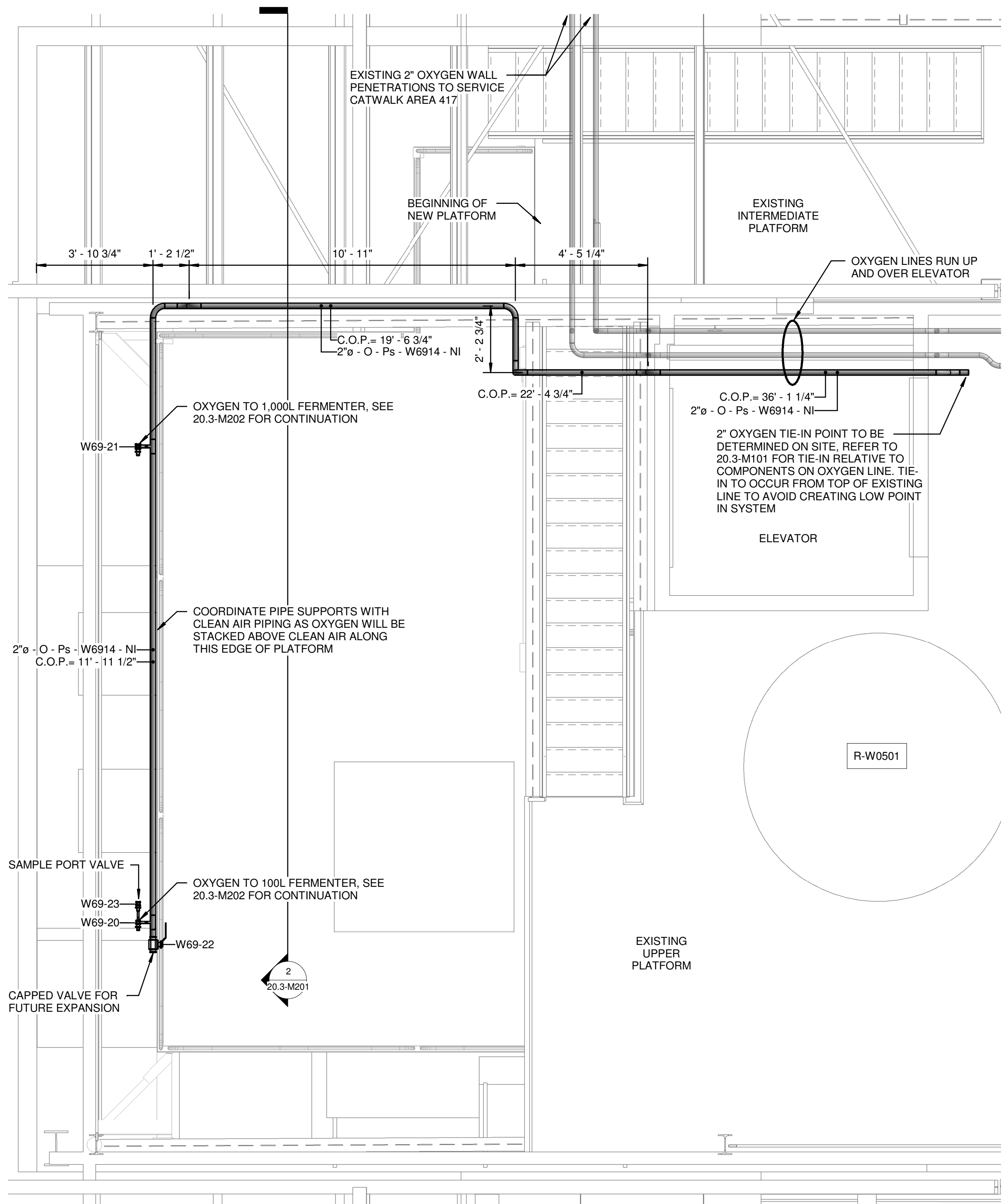
2 FERMENTATION - CLEAN AIR RUNOUTS ISOMETRIC

|                                  |          |                |           |
|----------------------------------|----------|----------------|-----------|
| Drawing Title:                   |          |                |           |
| FERMENTATION - CLEAN AIR RUNOUTS |          |                |           |
| Date:                            | 02/11/21 | Project No.:   | 21-515    |
| Scale:                           | AS NOTED | DWG No.:       | 20.2-M202 |
| Drawn:                           | KJG      | Origin Dwg:    |           |
| Checked:                         | SKR      | File No.:      | 21-515    |
|                                  |          | BioVectra Dwg: |           |



DRAWING NOTES:

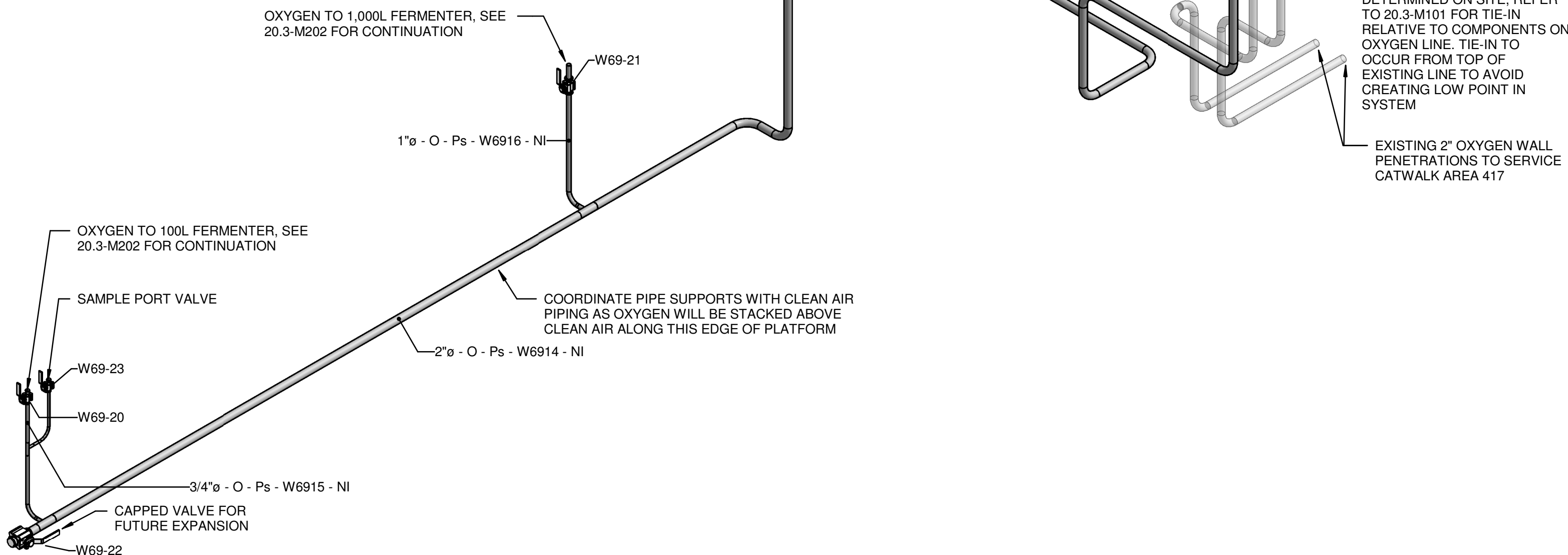
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- UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
- OXYGEN PIPING CLEANING TO BE DONE BY OWNER AFTER INSTALL.
- REFER TO DRAWING 20.3-M101 (PID-W69) FOR MORE DETAILS.



FERMENTATION - OXYGEN LAYOUT

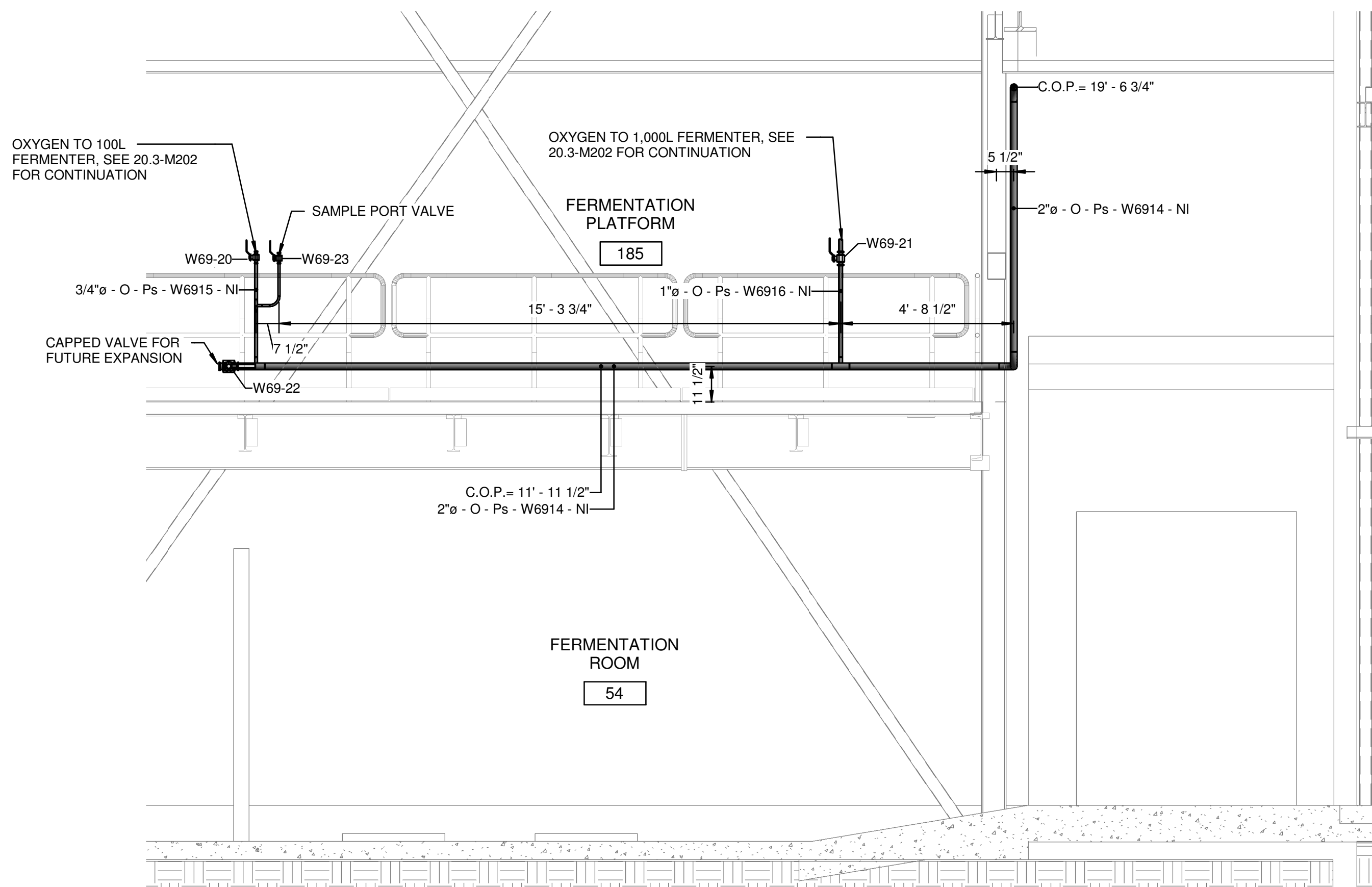
1

3/8" = 1'-0"



FERMENTATION - OXYGEN ISOMETRIC

3



FERMENTATION - OXYGEN SECTION

2

3/8" = 1'-0"

LEGEND:

|                    |                                      |
|--------------------|--------------------------------------|
| -----              | DEMOLITION PIPING                    |
| =====              | EXISTING PIPING                      |
| XX" - XX - XX      | NEW PIPING                           |
| -----              | PIPE TYPE (SEE SPECIFICATION)        |
| -----              | SERVICE (SEE SPECIFICATION)          |
| -----              | PIPE SIZE                            |
| C.O.P. = XX' - XX" | CENTER OF PIPE ELEVATION             |
| XX' - XX"          | POINT ELEVATION                      |
| XX' - XX"          | PIPE TYPE CHANGE (SEE SPECIFICATION) |
| X%                 | SLOPE INDICATOR                      |
| HP / LP            | HIGH POINT / LOW POINT               |

NOTES:

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- REFER TO PID LEGEND SHEET FOR LIST OF FLUID SERVICE DESIGNATIONS.
- ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).

|     |                         |          |
|-----|-------------------------|----------|
| 3   | ISSUED FOR CONSTRUCTION | 31-01-22 |
| 2   | ISSUED FOR 20-CCO-01    | 22-11-21 |
| 1   | ISSUED FOR PRICING      | 29-10-21 |
| NO. | Revision                | DD-MM-YY |

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- FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

Drawing Title:

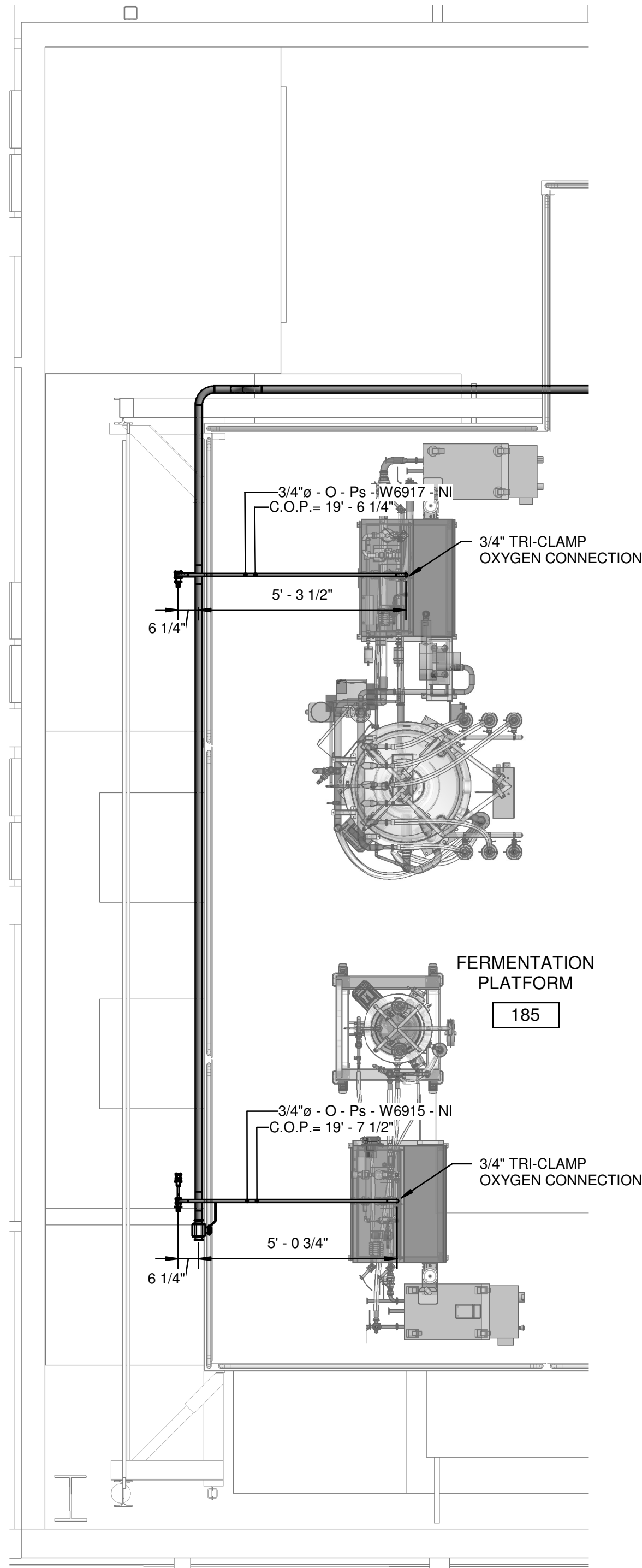
FERMENTATION - OXYGEN PLANS

|          |          |                |           |
|----------|----------|----------------|-----------|
| Date:    | 22/10/21 | Project No.:   | 21-515    |
| Scale:   | AS NOTED | DWG No.:       | 20.3-M201 |
| Drawn:   | KJG      | Origin Dwg:    |           |
| Checked: | SKR      | File No.:      | 21-515    |
|          |          | BioVectra Dwg: |           |



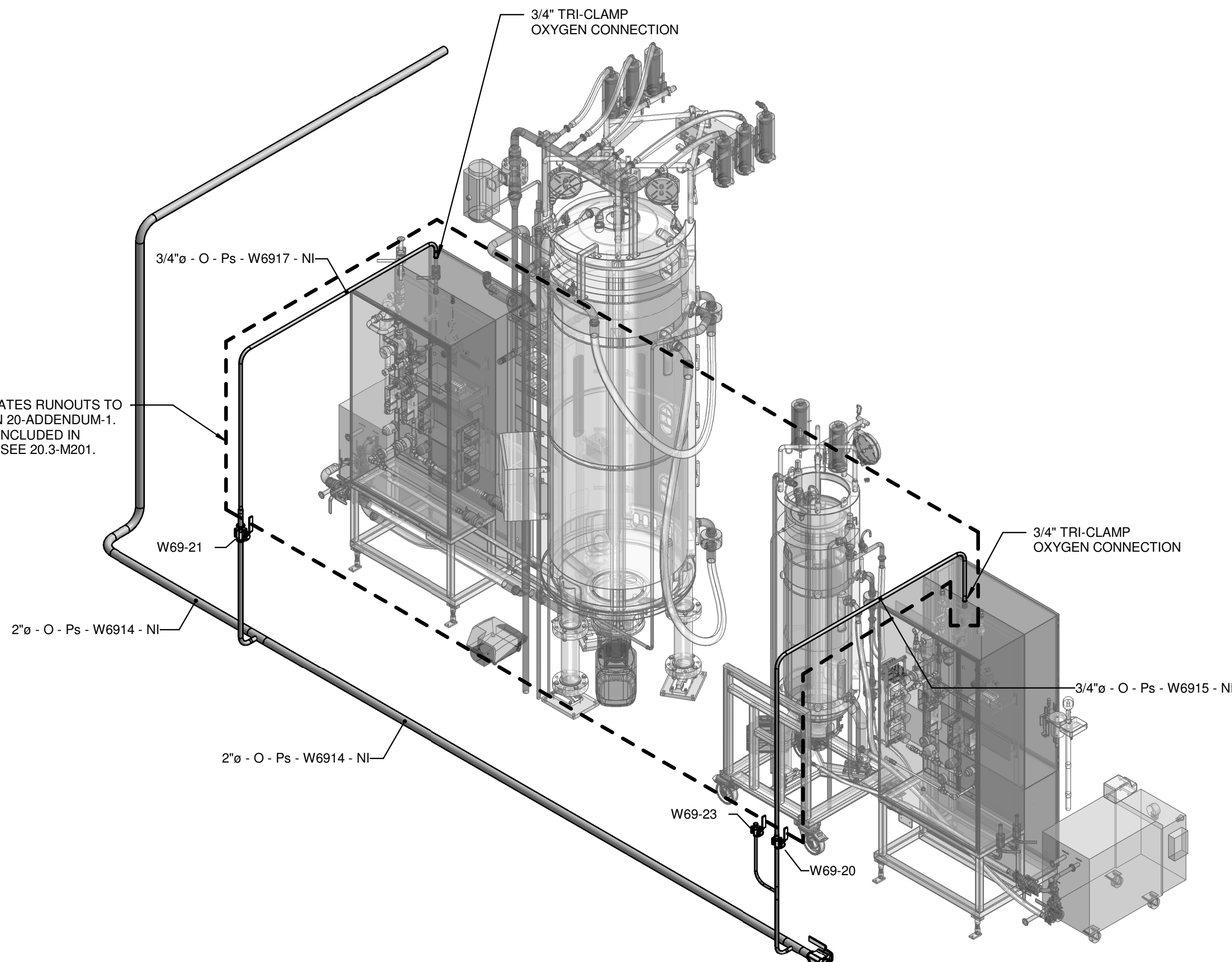
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- 6) UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
- 7) OXYGEN PIPING CLEANING TO BE DONE BY OWNER AFTER INSTALL.
- 8) REFER TO DRAWING 20.3-M101 (PID-W69) FOR MORE DETAILS.



1 FERMENTATION - OXYGEN RUNOUTS LAYOUT  
3/8" = 1'-0"

ENCLOSED PIPING INDICATES RUNOUTS TO EQUIPMENT INCLUDED IN 20-ADDENDUM-1. ALL OTHER PIPING WAS INCLUDED IN ORIGINAL BID PACKAGE, SEE 20.3-M201.



2 FERMENTATION - OXYGEN RUNOUTS ISOMETRIC

LEGEND:

|                    |                                      |
|--------------------|--------------------------------------|
| -----              | DEMOLITION PIPING                    |
| =====              | EXISTING PIPING                      |
| XX" - XX - XX      | NEW PIPING                           |
| ↑                  | PIPE TYPE (SEE SPECIFICATION)        |
| →                  | SERVICE (SEE SPECIFICATION)          |
| —                  | PIPE SIZE                            |
| C.O.P. = XX' - XX" | CENTER OF PIPE ELEVATION             |
| XX' - XX"          | POINT ELEVATION                      |
| XX ← XX            | PIPE TYPE CHANGE (SEE SPECIFICATION) |
| X%                 | SLOPE INDICATOR                      |
| HP / LP            | HIGH POINT / LOW POINT               |

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4. ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).

|   |                          |          |
|---|--------------------------|----------|
| 3 | ISSUED FOR CONSTRUCTION  | 31-01-22 |
| 2 | ISSUED FOR 20-CCO-01     | 22-11-21 |
| 1 | ISSUED FOR 20-ADDENDUM-1 | 09-11-21 |

| NO. | Revision | DD-MM-YY |
|-----|----------|----------|
|-----|----------|----------|

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4. FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

All measurements must be checked

Drawing Title:

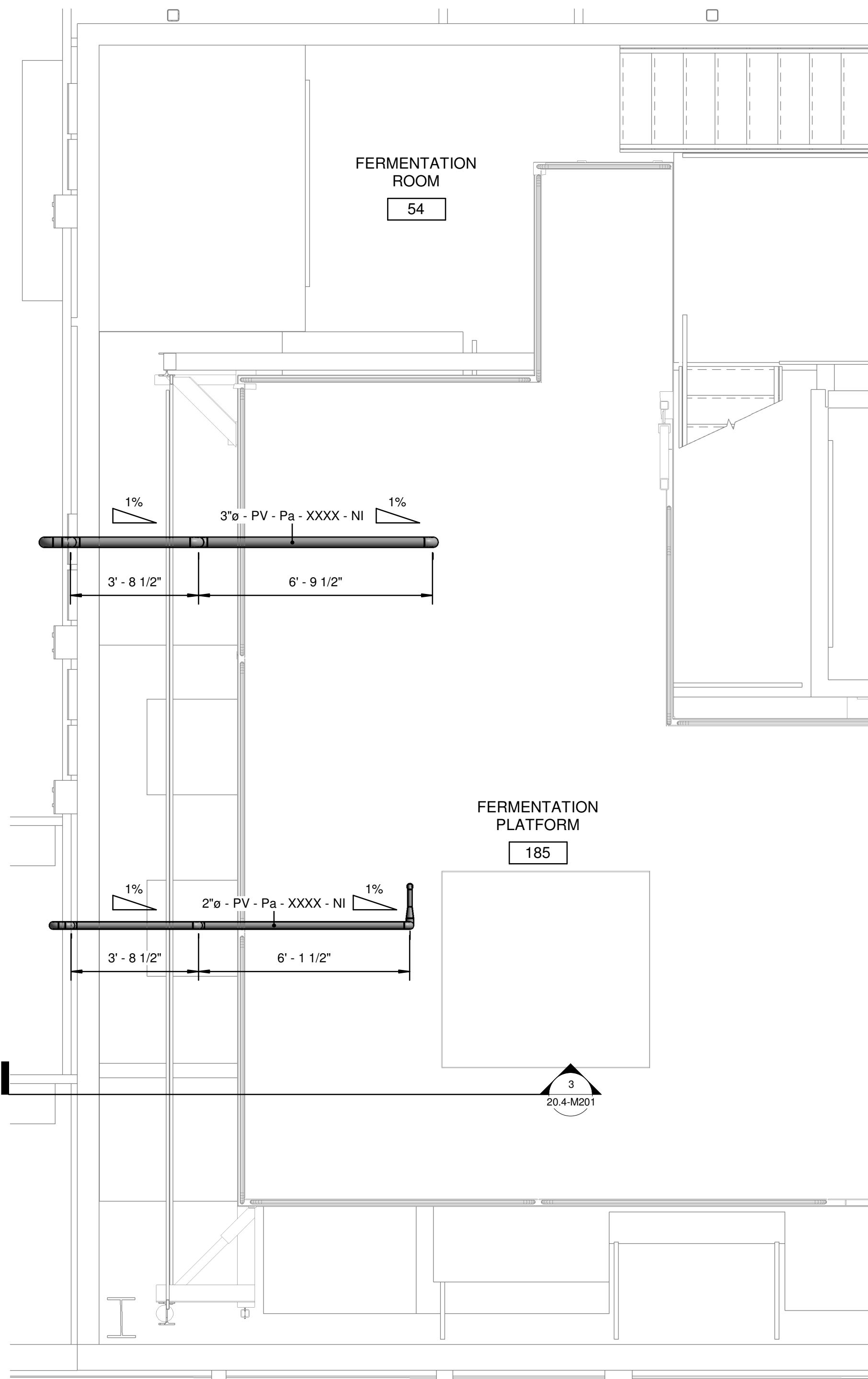
FERMENTATION - OXYGEN RUNOUTS

|           |          |                |           |
|-----------|----------|----------------|-----------|
| Date:     | 05/11/21 | Project No.:   | 21-515    |
| Scale:    | AS NOTED | DWG No.:       | 20.3-M202 |
| Drawn:    | KJG      | Origin Dwg:    |           |
| Checked:  | SKR      | Dwg:           |           |
| File No.: | 21-515   | BioVectra Dwg: |           |

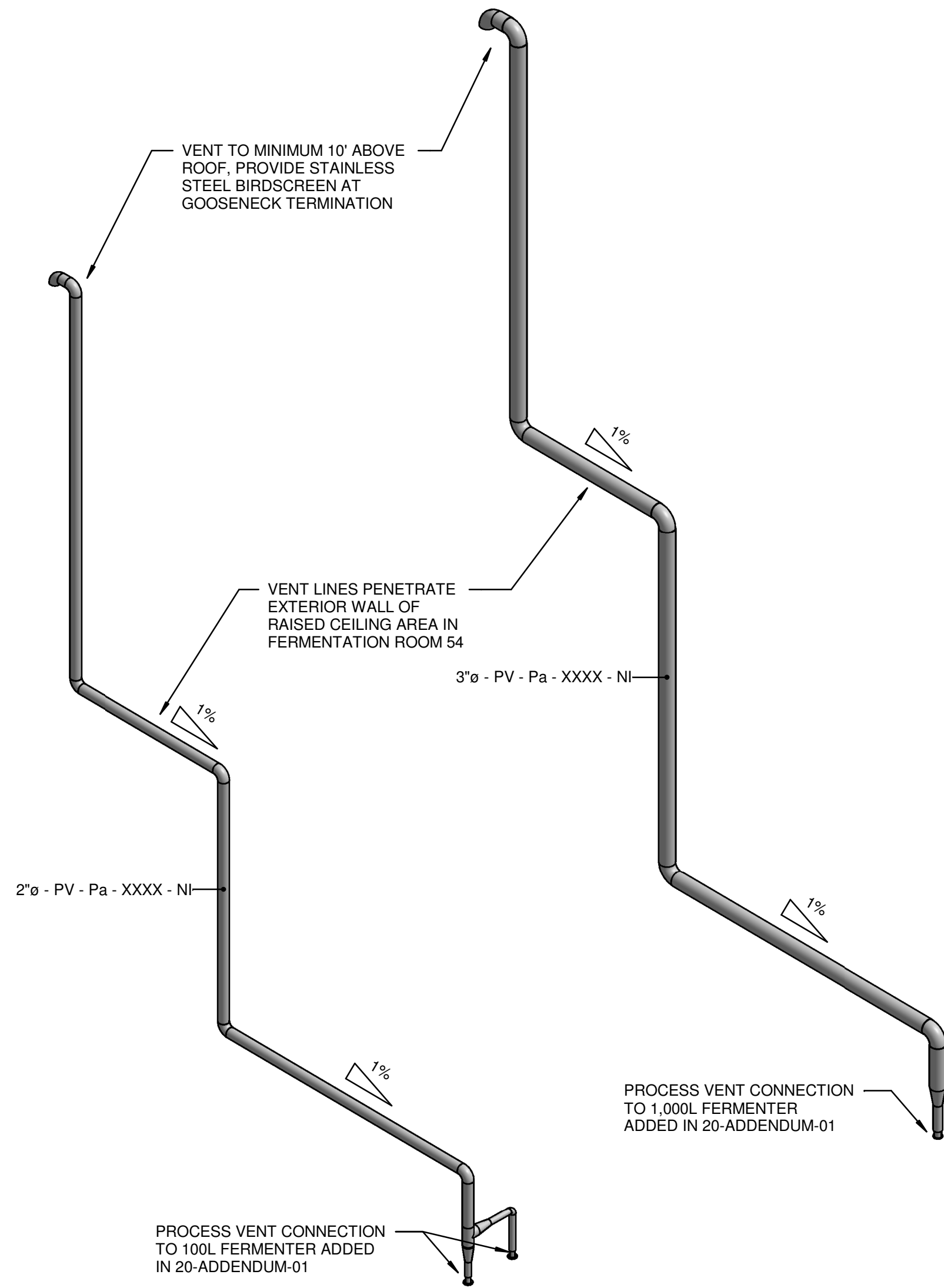


DRAWING NOTES:

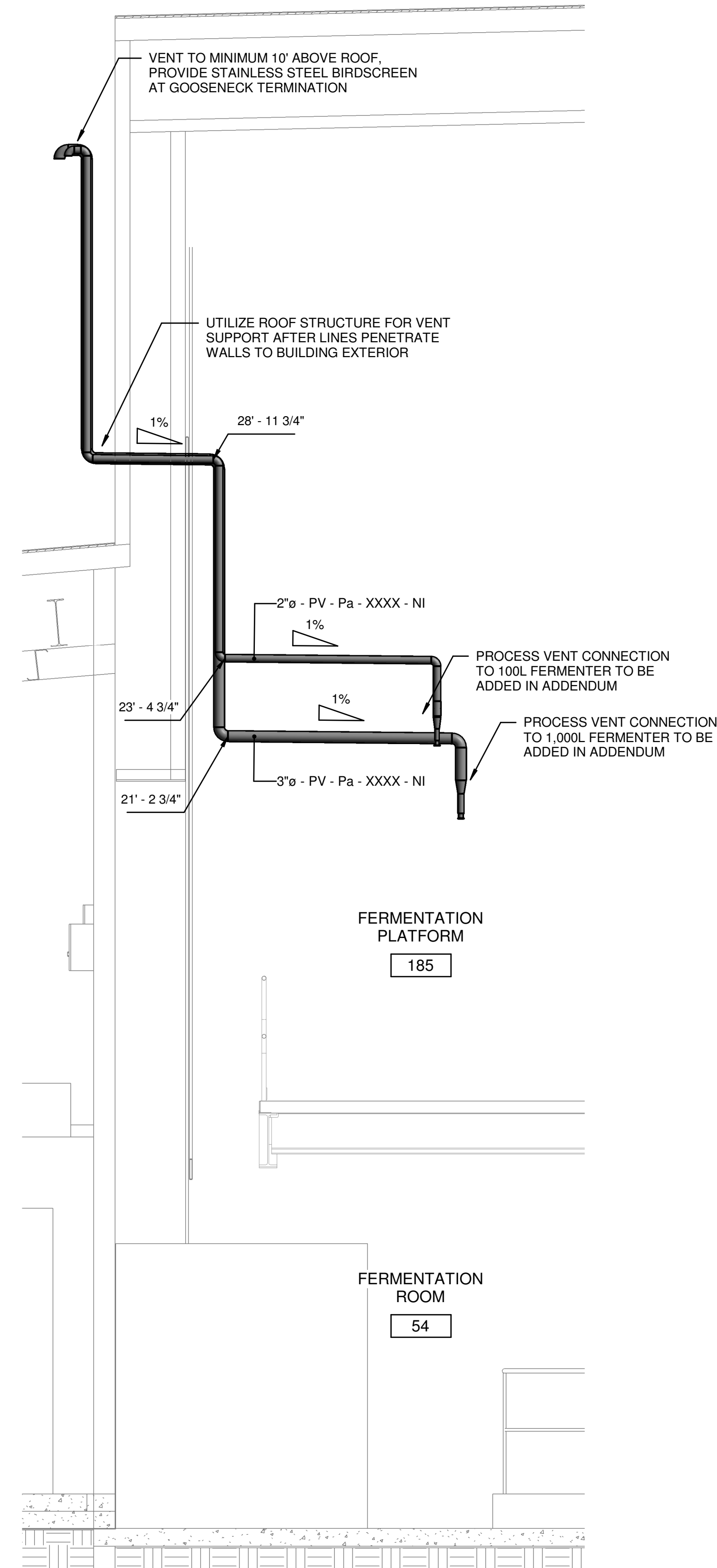
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- UTILIZE ROOF STRUCTURE FOR PROCESS VENT SUPPORT AFTER LINES PENETRATE WALLS TO BUILDING EXTERIOR.



1 FERMENTATION - PROCESS VENT LAYOUT  
3/8" = 1'-0"



2 FERMENTATION - PROCESS VENT ISOMETRIC  
3/8" = 1'-0"



3 FERMENTATION - PROCESS VENT SECTION  
3/8" = 1'-0"

LEGEND:

- DEMOLITION PIPING
- EXISTING PIPING
- XX" - XX - XX NEW PIPING
- PIPE TYPE (SEE SPECIFICATION)
- SERVICE (SEE SPECIFICATION)
- PIPE SIZE
- C.O.P.= XX' - XX" CENTER OF PIPE ELEVATION
- XX' - XX" POINT ELEVATION
- XX ← XX PIPE TYPE CHANGE (SEE SPECIFICATION)
- X% SLOPE INDICATOR
- HP / LP HIGH POINT / LOW POINT

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|     |                         |          |
|-----|-------------------------|----------|
| 2   | ISSUED FOR CONSTRUCTION | 31-01-22 |
| 1   | ISSUED FOR PRICING      | 29-10-21 |
| NO. | Revision                | DD-MM-YY |

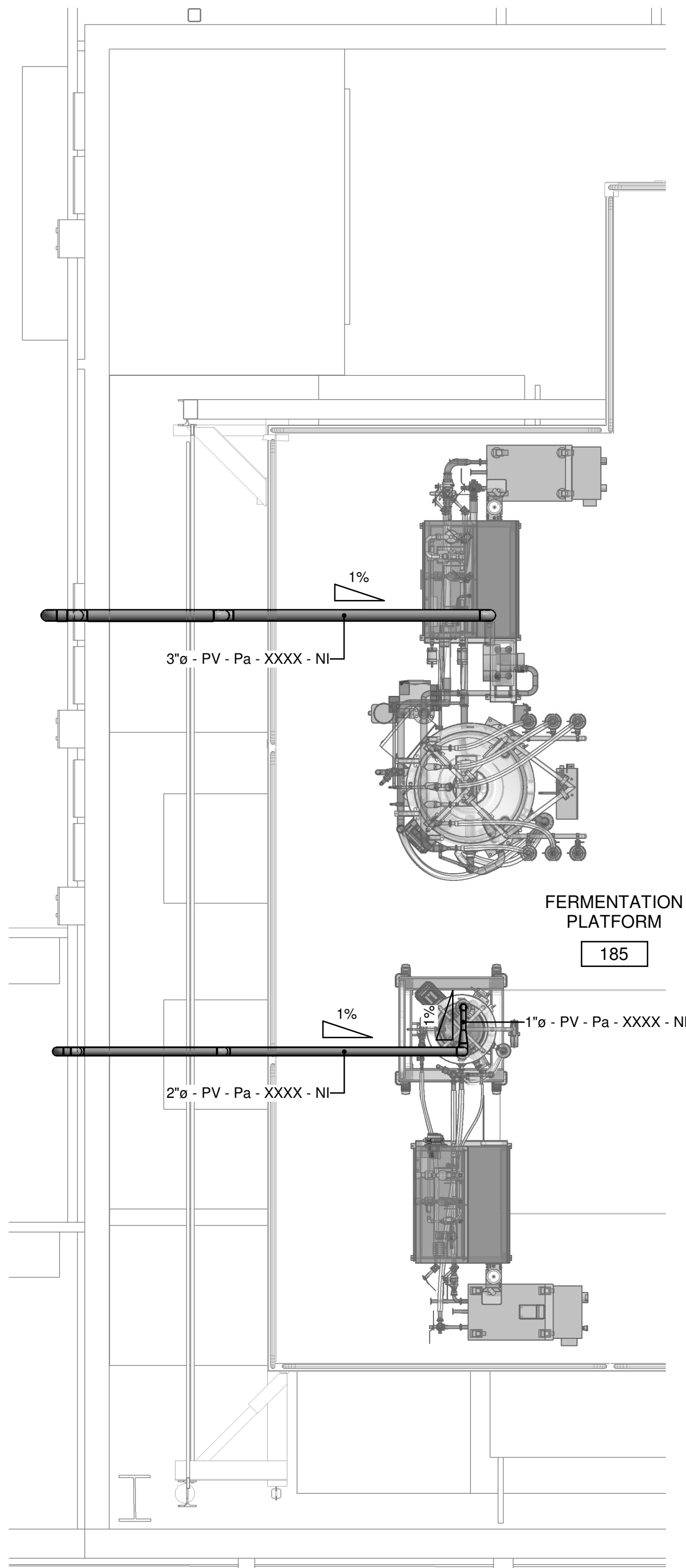
GENERAL NOTES:

- THIS DRAWING IS INTENDED FOR GENERAL ARRANGEMENT ONLY, NOT ALL EQUIPMENT IS SHOWN. REFER TO SCHEMATIC DRAWINGS, EQUIPMENT MANUFACTURERS INSTRUCTIONS, AND SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.
- HANGERS AND SUPPORTS WHERE INDICATED ARE FOR ILLUSTRATIVE PURPOSES. REFER TO SPECIFICATIONS FOR REQUIREMENTS.
- CONTRACTOR TO COORDINATE ALL ROUTING ON SITE PRIOR TO INSTALLATION. ANY ADJUSTMENTS REQUIRED TO ACCOMMODATE EXISTING CONDITIONS SHALL BE INCLUDED IN TENDER.
- FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

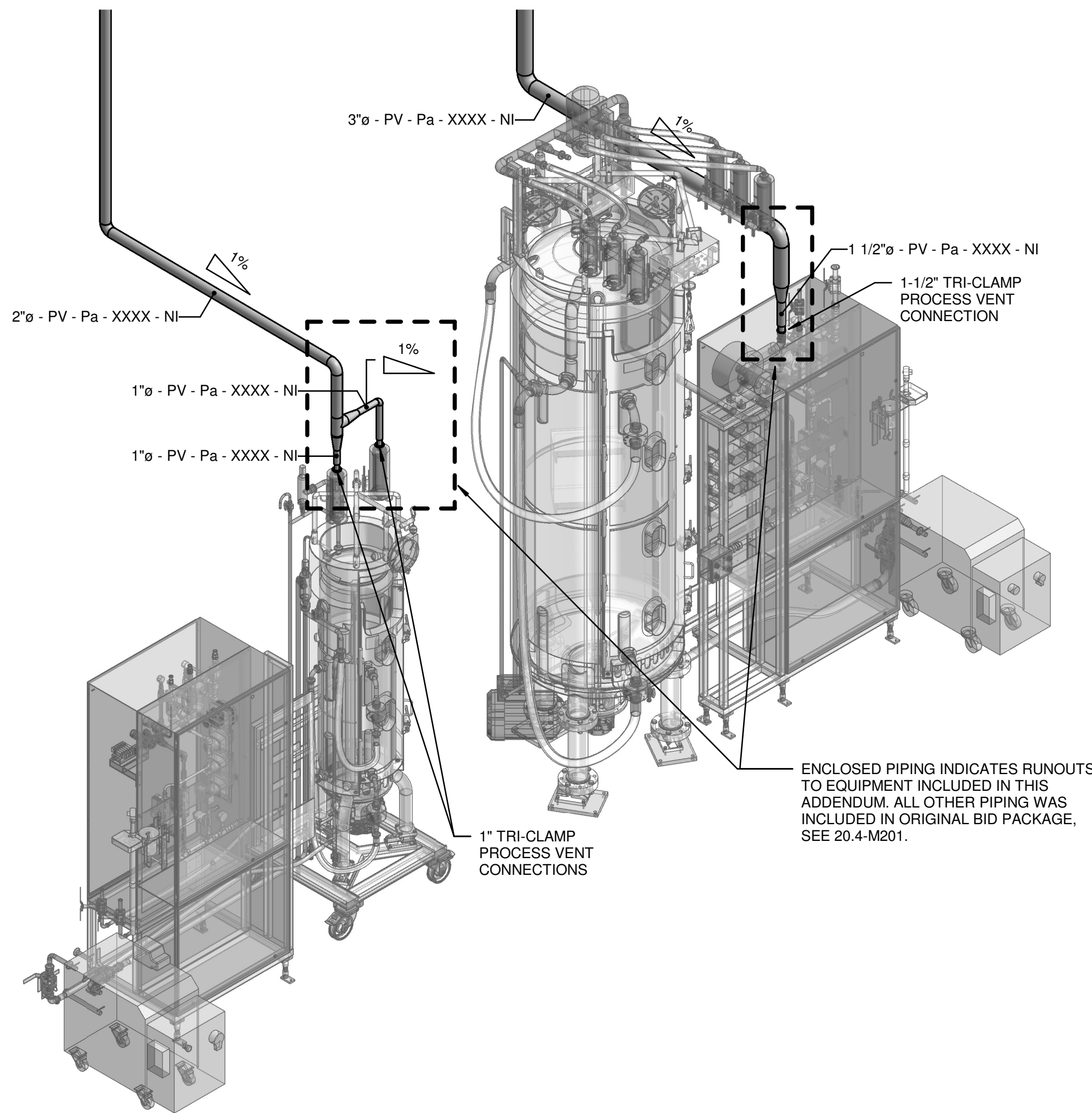
|                                   |          |                |           |
|-----------------------------------|----------|----------------|-----------|
| Drawing Title:                    |          |                |           |
| FERMENTATION - PROCESS VENT PLANS |          |                |           |
| Date:                             | 25/10/21 | Project No.:   | 21-515    |
| Scale:                            | AS NOTED | DWG No.:       | 20.4-M201 |
| Drawn:                            | KJG      | Origin Dwg:    |           |
| Checked:                          | SKR      | File No.:      | 21-515    |
|                                   |          | BioVectra Dwg: |           |



- DRAWING NOTES:**
- THIS DRAWING IS INTENDED FOR GENERAL ROUTE/LOCATIONS ONLY, COORDINATE EXACT ROUTE/LOCATIONS ON SITE.
  - TRANSLUCENT PIPING/EQUIPMENT REPRESENTS EXISTING PIPING THAT IS TO REMAIN.
  - SOLID PIPING/EQUIPMENT WITH BOLD LINES REPRESENTS NEW ELEMENTS.
  - COORDINATE WORK ON LINES WITH BIOVECTRA TO ENSURE LINES ARE DE-ENERGIZED WITH APPROPRIATE LOCK-OUT/TAG-OUT ENGAGED.
  - ALL SUPPORTS AND ASSOCIATED SUPPORT STRUCTURES ARE BY THIS CONTRACTOR. WHERE REQUIRED TO CREATE SUPPORT STRUCTURES FROM THE FERMENTATION PLATFORM, OR OTHERWISE, STAINLESS HSS SHALL BE USED. LOCATIONS OF SUPPORT STRUCTURE MUST BE COORDINATED WITH THE ENGINEER (MCA).
  - UTILIZE NEW PLATFORM STRUCTURE FOR PIPE SUPPORT WHERE POSSIBLE.
  - UTILIZE ROOF STRUCTURE FOR PROCESS VENT SUPPORT AFTER LINES PENETRATE WALLS TO BUILDING EXTERIOR.



1 FERMENTATION - PROCESS VENT RUNOUT LAYOUT  
3/8" = 1'-0"



2 FERMENTATION - PROCESS VENT RUNOUT ISOMETRIC

**LEGEND:**

- DEMOLITION PIPING  
----- EXISTING PIPING  
XX" - XX - XX NEW PIPING  
PIPE TYPE (SEE SPECIFICATION)  
SERVICE (SEE SPECIFICATION)  
PIPE SIZE  
C.O.P.= XX' - XX" CENTER OF PIPE ELEVATION  
XX' - XX" POINT ELEVATION  
XX' - XX" PIPE TYPE CHANGE (SEE SPECIFICATION)  
X% SLOPE INDICATOR  
HP / LP HIGH POINT / LOW POINT

**NOTES:**

- THE WRITTEN SPECIFICATIONS FORM AN INTEGRAL PART OF THE CONTRACT DOCUMENTS AND CONTAIN IMPORTANT INFORMATION ON EXECUTION OF WORK. REVIEW IN DETAIL. SECTIONS OF NOTE WITH SPECIFIC REQUIREMENTS INCLUDE BUT ARE NOT LIMITED TO SCOPES OF WORK, COORDINATION, CLEANING, IOM AND TURN-OVER DOCUMENTATION REQUIREMENTS, AND FABRICATION/INSTALLATION REQUIREMENTS.
- DIMENSIONS NOTED ON LAYOUT ARE TO CENTER OF PIPE.
- REFER TO PID LEGEND SHEET FOR LIST OF FLUID SERVICE DESIGNATIONS.
- ALL ELEVATIONS ARE MEASURED FROM THE FINISHED FLOOR (GROUND LEVEL 0'-0" AT LOWER ELEVATOR ENTRANCE).

|     |                          |          |
|-----|--------------------------|----------|
| 2   | ISSUED FOR CONSTRUCTION  | 31-01-22 |
| 1   | ISSUED FOR 20-ADDENDUM-1 | 09-11-21 |
| NO. | Revision                 | DD-MM-YY |

**GENERAL NOTES:**

- THIS DRAWING IS INTENDED FOR GENERAL ARRANGEMENT ONLY, NOT ALL EQUIPMENT IS SHOWN. REFER TO SCHEMATIC DRAWINGS, EQUIPMENT MANUFACTURERS INSTRUCTIONS, AND SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.
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- CONTRACTOR TO COORDINATE ALL ROUTING ON SITE PRIOR TO INSTALLATION. ANY ADJUSTMENTS REQUIRED TO ACCOMMODATE EXISTING CONDITIONS SHALL BE INCLUDED IN TENDER.
- FIRE STOP PIPING AT ALL RATED FIRE SEPARATIONS.

Drawing Title:

FERMENTATION - PROCESS VENT RUNOUTS

|           |          |                |           |
|-----------|----------|----------------|-----------|
| Date:     | 08/11/21 | Project No.:   | 21-515    |
| Scale:    | AS NOTED | DWG No.:       | 20.4-M202 |
| Drawn:    | KJG      | Origin Dwg:    |           |
| Checked:  | SKR      | Dwg:           |           |
| File No.: | 21-515   | BioVectra Dwg: |           |

